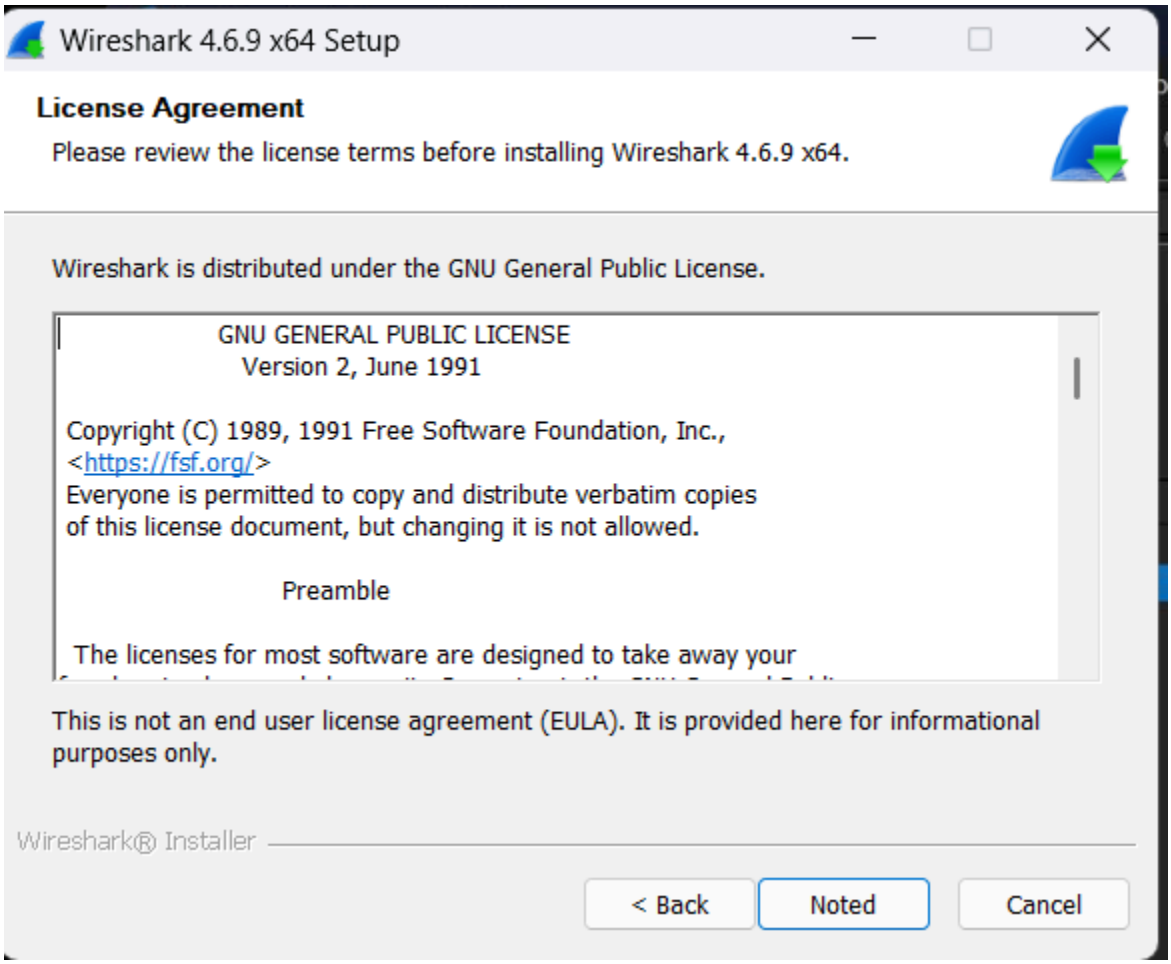
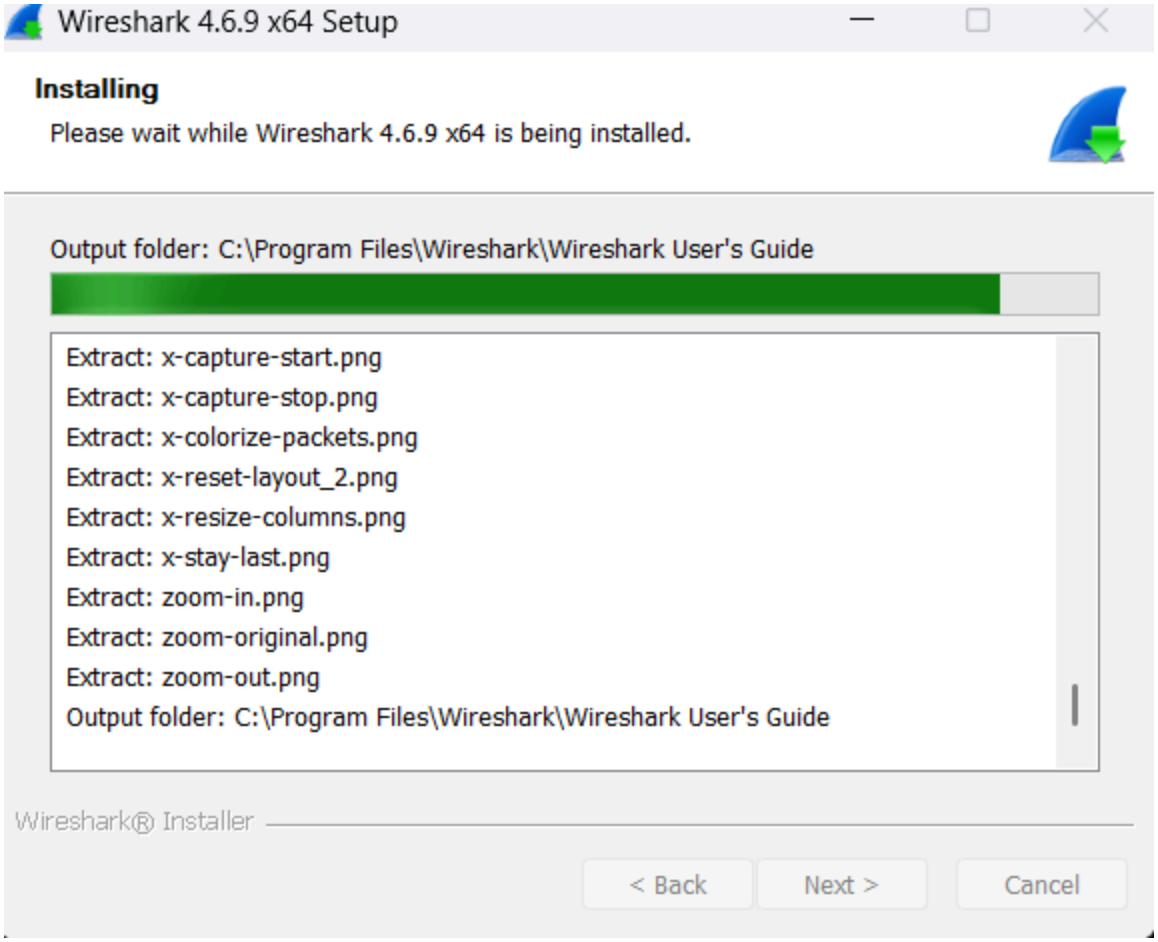
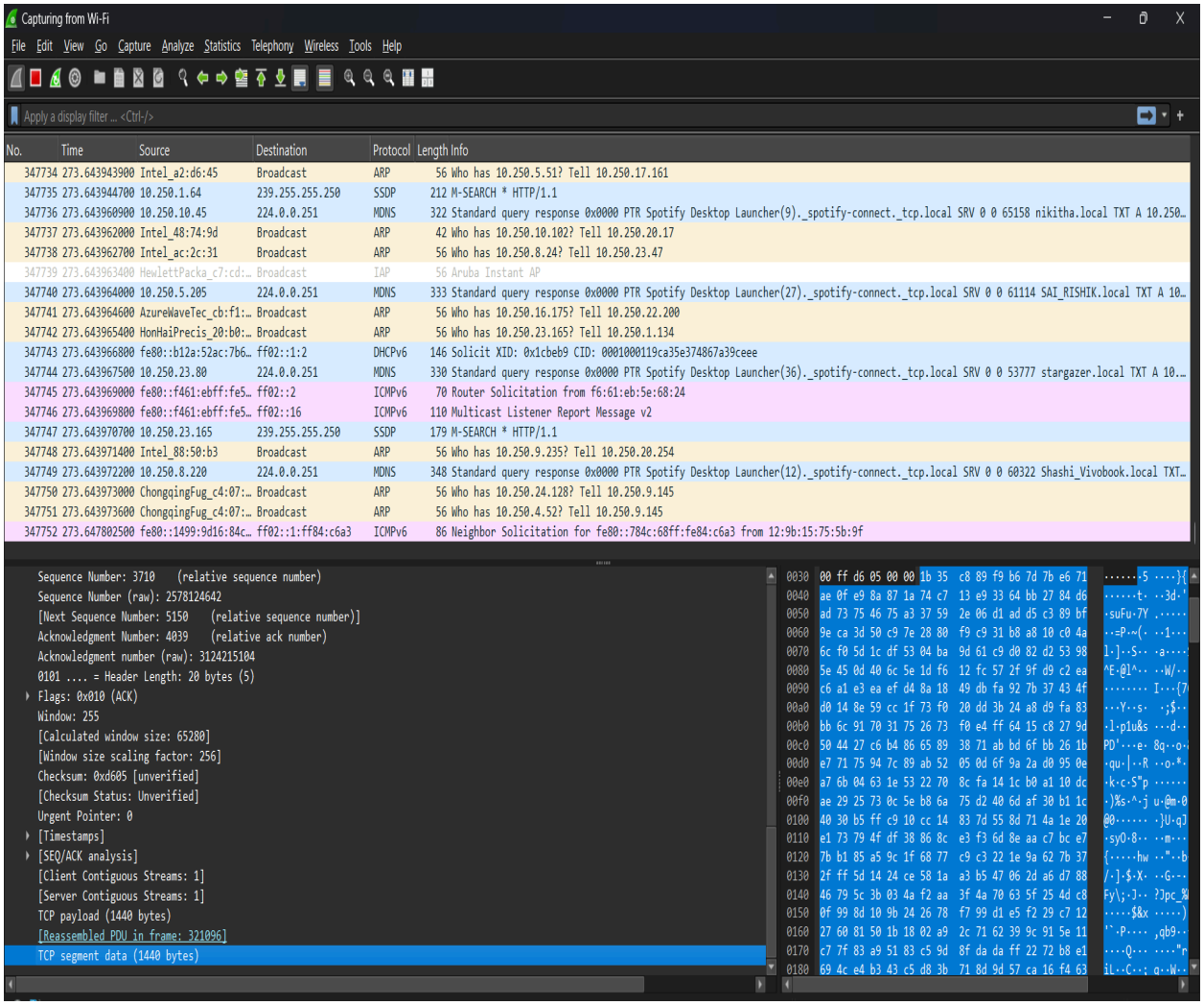
Skill - 1

Install Wireshark



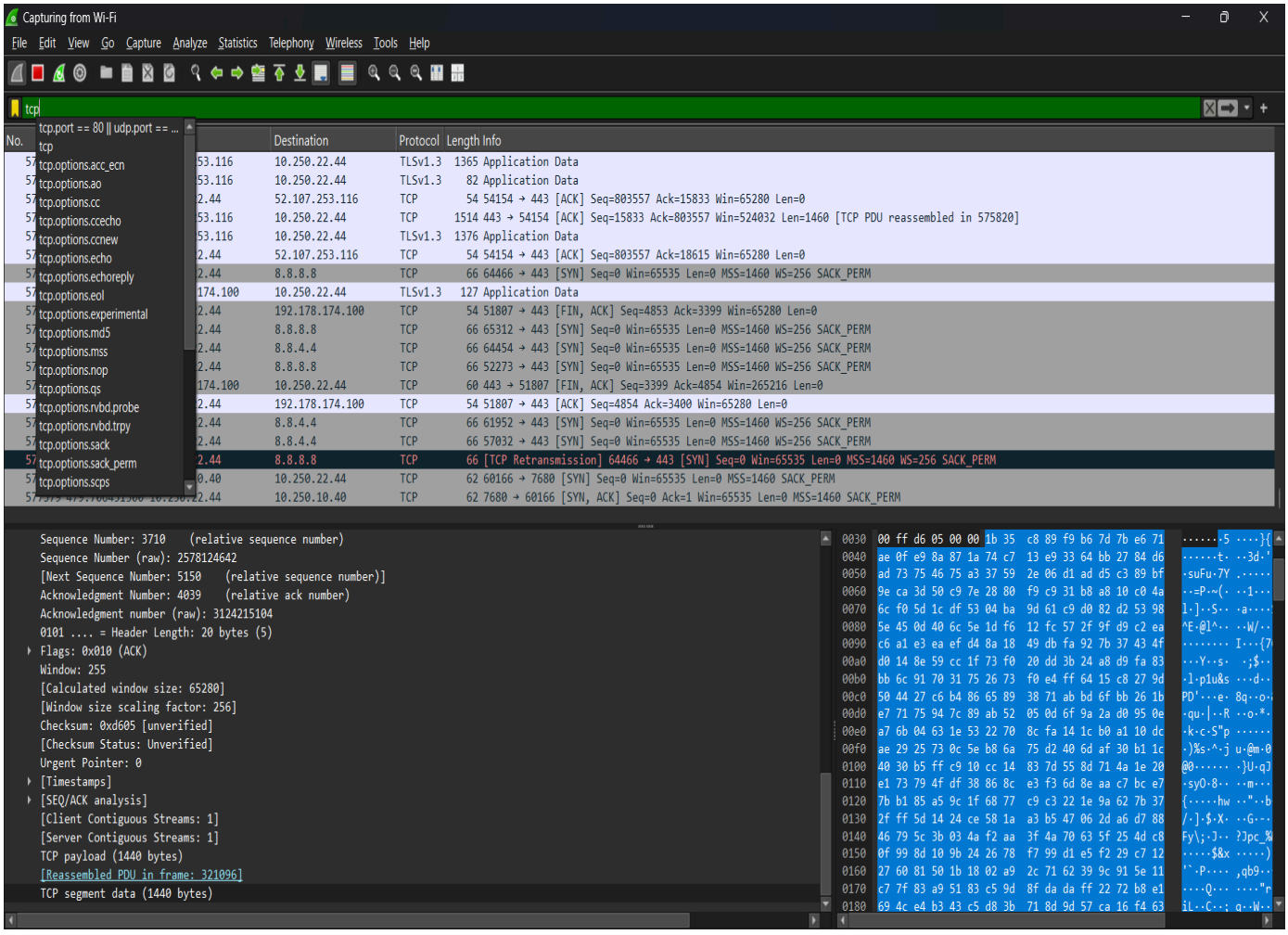


Generate network traffic

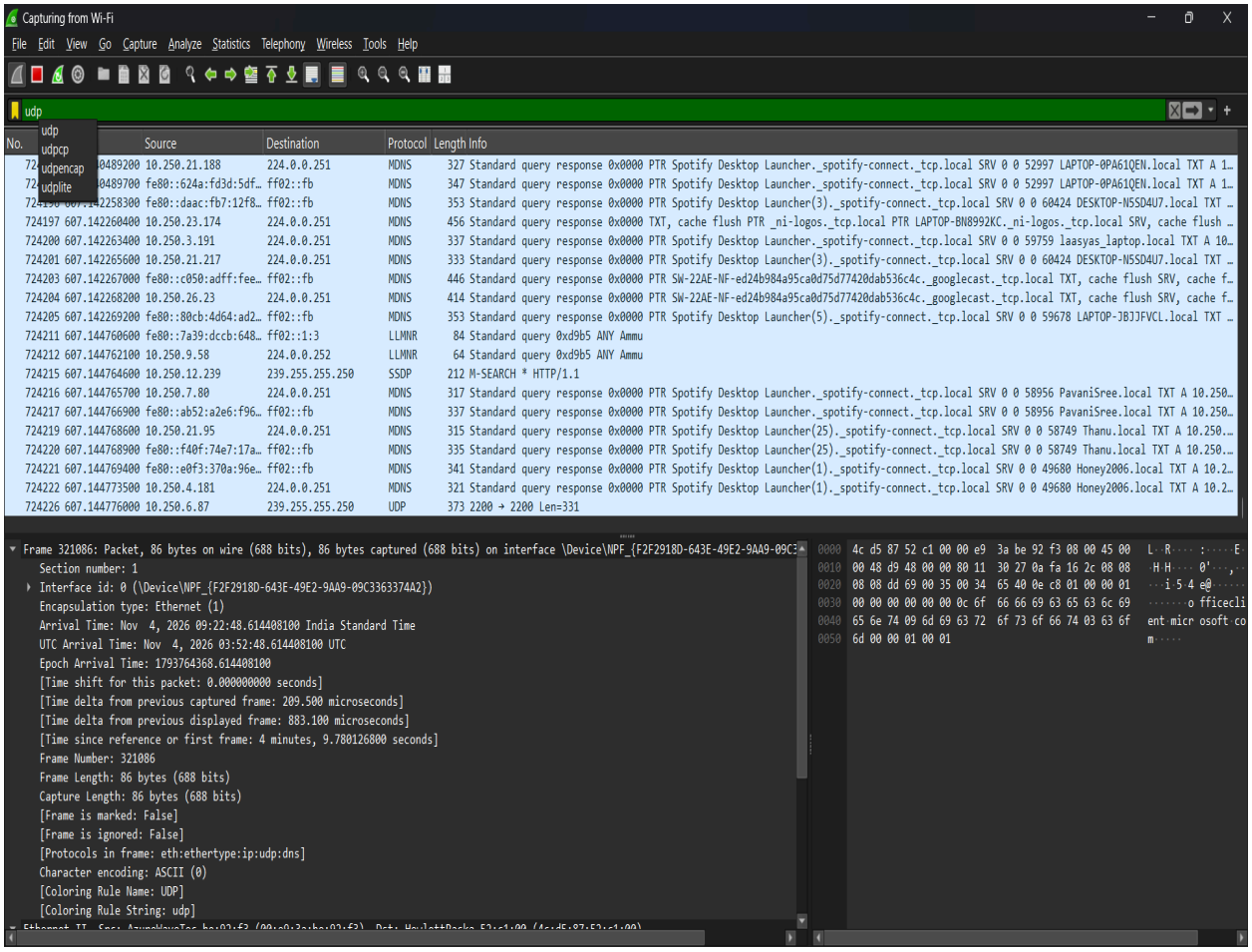


Apply Display Filter

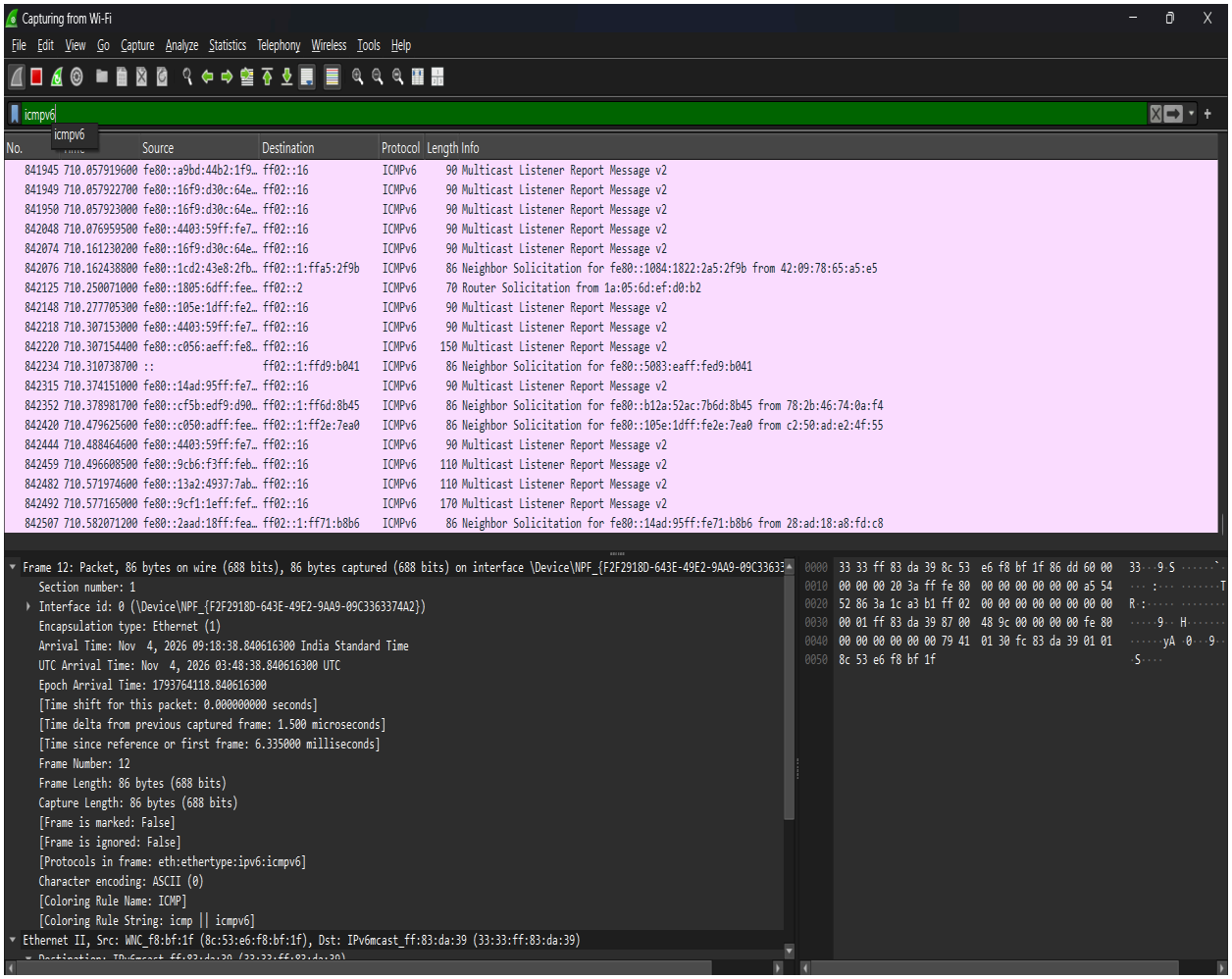
1.tcp


Capturing from Wi-Fi
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
tcp
tcp.port == 80 || udp.port == ...
tcp
tcp.options.acc_ecn
tcp.options.ao
tcp.options.cc
tcp.options.ccecho
tcp.options.ccnew
tcp.options.echo
tcp.options.echoreply
tcp.options.eol
tcp.options.experimental
tcp.options.md5
tcp.options.mss
tcp.options.nop
tcp.options.qs
tcp.options.rvbd.probe
tcp.options.rvbd.trpy
tcp.options.sack
tcp.options.sack_perm
tcp.options.scps
No. Destination Protocol Length Info
10.250.22.44 TLSv1.3 1365 Application Data
10.250.22.44 TLSv1.3 82 Application Data
52.107.253.116 TCP 54 54154 → 443 [ACK] Seq=803557 Ack=15833 Win=65280 Len=0
10.250.22.44 TCP 1514 443 → 54154 [ACK] Seq=15833 Ack=803557 Win=524032 Len=1460 [TCP PDU reassembled in 575820]
10.250.22.44 TLSv1.3 1376 Application Data
52.107.253.116 TCP 54 54154 → 443 [ACK] Seq=803557 Ack=18615 Win=65280 Len=0
8.8.8.8 TCP 66 64466 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
10.250.22.44 TLSv1.3 127 Application Data
192.178.174.100 TCP 54 51807 → 443 [FIN, ACK] Seq=4853 Ack=3399 Win=65280 Len=0
8.8.8.8 TCP 66 65312 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
8.8.4.4 TCP 66 64454 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
8.8.8.8 TCP 66 52273 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
10.250.22.44 TCP 60 443 → 51807 [FIN, ACK] Seq=3399 Ack=4854 Win=265216 Len=0
192.178.174.100 TCP 54 51807 → 443 [ACK] Seq=4854 Ack=3400 Win=65280 Len=0
8.8.4.4 TCP 66 61952 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
8.8.4.4 TCP 66 57032 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
8.8.8.8 TCP 66 [TCP Retransmission] 64466 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
10.250.22.44 TCP 62 60166 → 7680 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 SACK_PERM
10.250.10.40 TCP 62 7680 → 60166 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1460 SACK_PERM
Sequence Number: 3710 (relative sequence number)
Sequence Number (raw): 2578124642
[Next Sequence Number: 5150 (relative sequence number)]
Acknowledgment Number: 4039 (relative ack number)
Acknowledgment number (raw): 3124215104
0101 .... = Header Length: 20 bytes (5)
Flags: 0x010 (ACK)
Window: 255
[Calculated window size: 65280]
[Window size scaling factor: 256]
Checksum: 0xd605 [unverified]
[Checksum Status: Unverified]
Urgent Pointer: 0
[Timestamps]
[SEQ/ACK analysis]
[Client Contiguous Streams: 1]
[Server Contiguous Streams: 1]
TCP payload (1440 bytes)
[Reassembled PDU in frame: 321096]
TCP segment data (1440 bytes)
0030 00 ff d6 05 00 00 1b 35 c8 89 f9 b6 7d 7b e6 71
0040 ae 0f e9 8a 87 1a 74 c7 13 e9 33 64 bb 27 84 d6
0050 ad 73 75 46 75 a3 37 59 2e 06 d1 ad d5 c3 89 bf
0060 9e ca 3d 50 c9 7e 28 80 f9 c9 31 b8 a8 10 c0 4a
0070 6c f0 5d 1c df 53 04 ba 9d 61 c9 d0 82 d2 53 98
0080 5e 45 0d 40 6c 5e 1d f6 12 fc 57 2f 9f d9 c2 ea
0090 c6 a1 e3 ea ef d4 8a 18 49 db fa 92 7b 37 43 4f
00a0 d0 14 8e 59 cc 1f 73 f0 20 dd 3b 24 a8 d9 fa 83
00b0 bb 6c 91 70 31 75 26 73 f0 e4 ff 64 15 c8 27 9d
00c0 50 44 27 c6 b4 86 65 89 38 71 ab bd 6f bb 26 1b
00d0 e7 71 75 94 7c 89 ab 52 05 0d 6f 9a 2a d0 95 0e
00e0 a7 6b 04 63 1e 53 22 70 8c fa 14 1c b0 a1 10 dc
00f0 ae 29 25 73 0c 5e b8 6a 75 d2 40 6d af 30 b1 1c
0100 40 30 b5 ff c9 10 cc 14 83 7d 55 8d 71 4a 1e 20
0110 e1 73 79 4f df 38 86 8c e3 f3 6d 8e aa c7 bc e7
0120 7b b1 85 a5 9c 1f 68 77 c9 c3 22 1e 9a 62 7b 37
0130 2f ff 5d 14 24 ce 58 1a a3 b5 47 06 2d a6 d7 88
0140 46 79 5c 3b 03 4a f2 aa 3f 4a 70 63 5f 25 4d c8
0150 0f 99 8d 10 9b 24 26 78 f7 99 d1 e5 f2 29 c7 12
0160 27 60 81 50 1b 18 02 a9 2c 71 62 39 9c 91 5e 11
0170 c7 7f 83 a9 51 83 c5 9d 8f da da ff 22 72 b8 e1
0180 69 4c e4 b3 43 c5 d8 3b 71 8d 9d 57 ca 16 f4 63

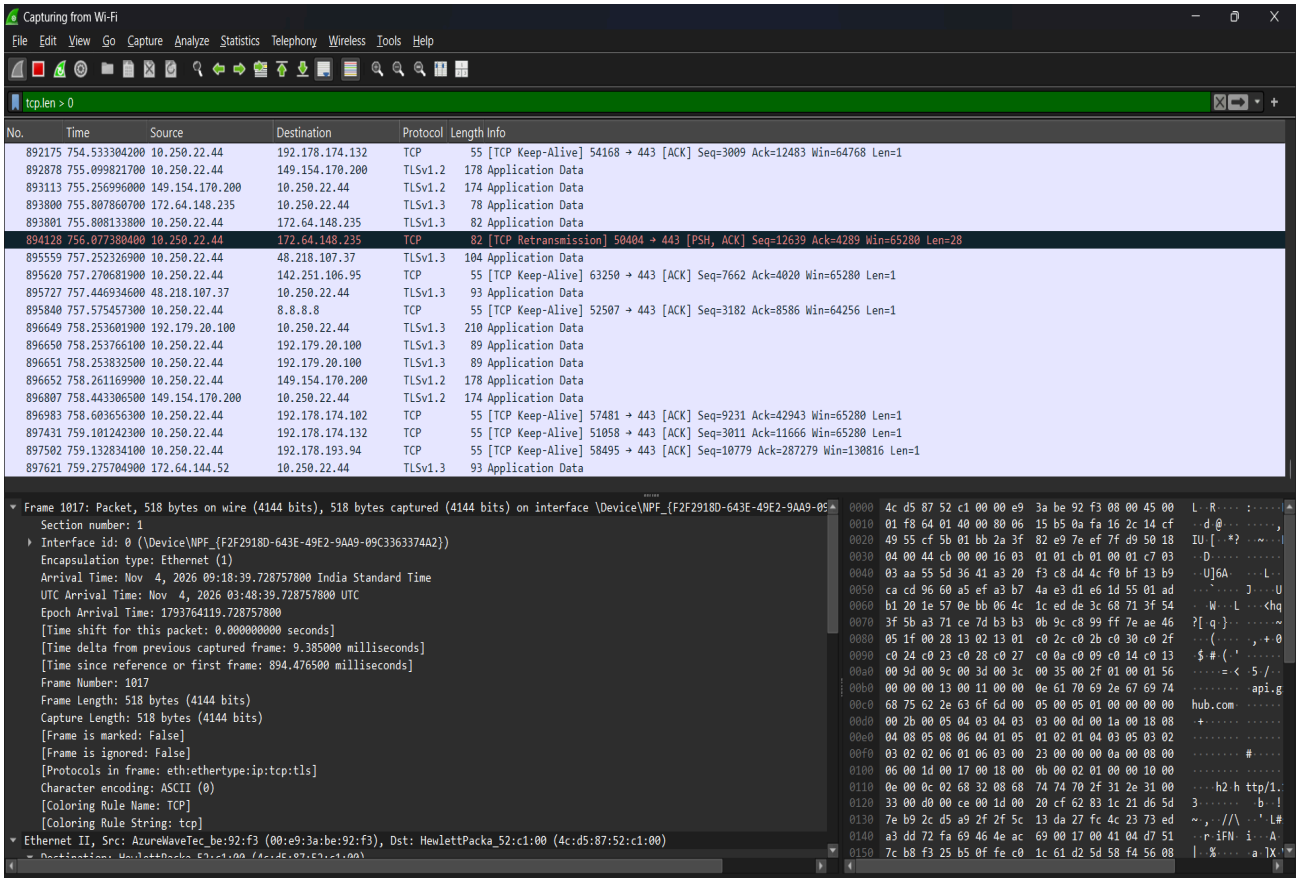
2.udp


Capturing from Wi-Fi
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
udp
udp
udpcp
udpencap
udplite
No. Source Destination Protocol Length Info
724197 607.142260400 10.250.23.174 224.0.0.251 MDNS 456 Standard query response 0x0000 TXT, cache flush PTR _ni-logos._tcp.local PTR LAPTOP-BN8992KC._ni-logos._tcp.local SRV, cache flush
724200 607.142263400 10.250.3.191 224.0.0.251 MDNS 337 Standard query response 0x0000 PTR Spotify Desktop Launcher._spotify-connect._tcp.local SRV 0 0 59759 laasyas_laptop.local TXT A 10
724201 607.142265600 10.250.21.217 224.0.0.251 MDNS 333 Standard query response 0x0000 PTR Spotify Desktop Launcher(3)._spotify-connect._tcp.local SRV 0 0 60424 DESKTOP-N5SD4U7.local TXT
724203 607.142267000 fe80::c050:adff:fee… ff02::fb MDNS 446 Standard query response 0x0000 PTR SW-22AE-NF-ed24b984a95ca0d75d77420dab536c4c._googlecast._tcp.local TXT, cache flush SRV, cache f
724204 607.142268200 10.250.26.23 224.0.0.251 MDNS 414 Standard query response 0x0000 PTR SW-22AE-NF-ed24b984a95ca0d75d77420dab536c4c._googlecast._tcp.local TXT, cache flush SRV, cache f
724205 607.142269200 fe80::80cb:4d64:ad2… ff02::fb MDNS 353 Standard query response 0x0000 PTR Spotify Desktop Launcher(5)._spotify-connect._tcp.local SRV 0 0 59678 LAPTOP-JBJJFVCL.local TXT
724211 607.144760600 fe80::7a39:dccb:648… ff02::1:3 LLMNR 84 Standard query 0xd9b5 ANY Ammu
724212 607.144762100 10.250.9.58 224.0.0.252 LLMNR 64 Standard query 0xd9b5 ANY Ammu
724215 607.144764600 10.250.12.239 239.255.255.250 SSDP 212 M-SEARCH * HTTP/1.1
724216 607.144765700 10.250.7.80 224.0.0.251 MDNS 317 Standard query response 0x0000 PTR Spotify Desktop Launcher._spotify-connect._tcp.local SRV 0 0 58956 PavaniSree.local TXT A 10.250
724217 607.144766900 fe80::ab52:a2e6:f96… ff02::fb MDNS 337 Standard query response 0x0000 PTR Spotify Desktop Launcher._spotify-connect._tcp.local SRV 0 0 58956 PavaniSree.local TXT A 10.250
724219 607.144768600 10.250.21.95 224.0.0.251 MDNS 315 Standard query response 0x0000 PTR Spotify Desktop Launcher(25)._spotify-connect._tcp.local SRV 0 0 58749 Thanu.local TXT A 10.250
724220 607.144768900 fe80::f40f:74e7:17a… ff02::fb MDNS 335 Standard query response 0x0000 PTR Spotify Desktop Launcher(25)._spotify-connect._tcp.local SRV 0 0 58749 Thanu.local TXT A 10.250
724221 607.144769400 fe80::e0f3:370a:96e… ff02::fb MDNS 341 Standard query response 0x0000 PTR Spotify Desktop Launcher(1)._spotify-connect._tcp.local SRV 0 0 49680 Honey2006.local TXT A 10.2
724222 607.144773500 10.250.4.181 224.0.0.251 MDNS 321 Standard query response 0x0000 PTR Spotify Desktop Launcher(1)._spotify-connect._tcp.local SRV 0 0 49680 Honey2006.local TXT A 10.2
724226 607.144776000 10.250.6.87 239.255.255.250 UDP 373 2200 → 2200 Len=331
Frame 321086: Packet, 86 bytes on wire (688 bits), 86 bytes captured (688 bits) on interface \Device\NPF_{F2F2918D-643E-49E2-9AA9-09C3
Section number: 1
Interface id: 0 (\Device\NPF_{F2F2918D-643E-49E2-9AA9-09C3363374A2})
Encapsulation type: Ethernet (1)
Arrival Time: Nov 4, 2026 09:22:48.614408100 India Standard Time
UTC Arrival Time: Nov 4, 2026 03:52:48.614408100 UTC
Epoch Arrival Time: 1793764368.614408100
[Time shift for this packet: 0.000000000 seconds]
[Time delta from previous captured frame: 209.500 microseconds]
[Time delta from previous displayed frame: 883.100 microseconds]
[Time since reference or first frame: 4 minutes, 9.780126800 seconds]
Frame Number: 321086
Frame Length: 86 bytes (688 bits)
Capture Length: 86 bytes (688 bits)
[Frame is marked: False]
[Frame is ignored: False]
[Protocols in frame: eth:ethertype:ip:udp:dns]
Character encoding: ASCII (0)
[Coloring Rule Name: UDP]
[Coloring Rule String: udp]
0000 4c d5 87 52 c1 00 00 e9 3a be 92 f3 08 00 45 00
0010 00 48 d9 48 00 00 80 11 30 27 0a fa 16 2c 08 08
0020 08 08 dd 69 00 35 00 34 65 40 0e c8 01 00 00 01
0030 00 00 00 00 00 00 0c 6f 66 66 69 63 65 63 6c 69
0040 65 6e 74 09 6d 69 63 72 6f 73 6f 66 74 03 63 6f
0050 6d 00 00 01 00 01

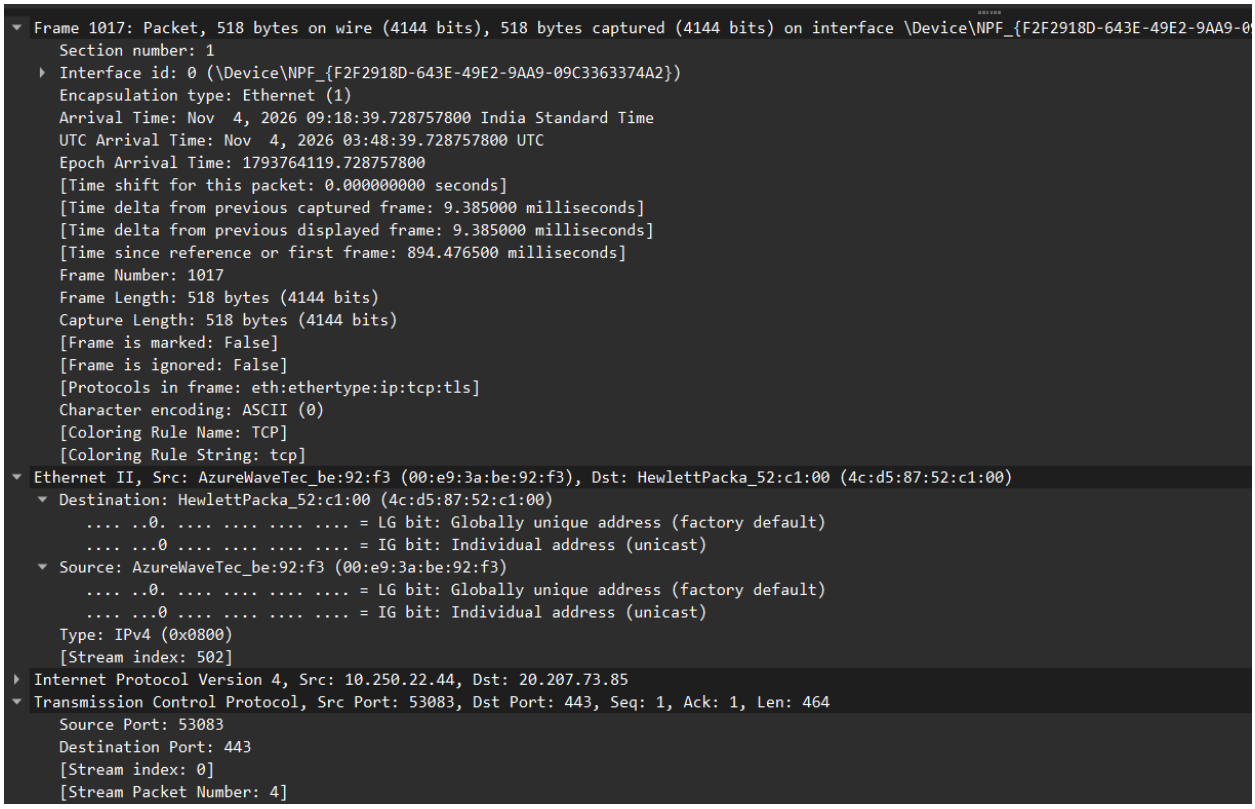
3.icmpv6


Capturing from Wi-Fi
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
icmpv6
No. Time Source Destination Protocol Length Info
841945 710.057919600 fe80::a9bd:44b2:1f9… ff02::16 ICMPv6 90 Multicast Listener Report Message v2
841949 710.057922700 fe80::16f9:d30c:64e… ff02::16 ICMPv6 90 Multicast Listener Report Message v2
841950 710.057923000 fe80::16f9:d30c:64e… ff02::16 ICMPv6 90 Multicast Listener Report Message v2
842048 710.076959500 fe80::4403:59ff:fe7… ff02::16 ICMPv6 90 Multicast Listener Report Message v2
842074 710.161230200 fe80::16f9:d30c:64e… ff02::16 ICMPv6 90 Multicast Listener Report Message v2
842076 710.162438800 fe80::1cd2:43e8:2fb… ff02::1:ffa5:2f9b ICMPv6 86 Neighbor Solicitation for fe80::1084:1822:2a5:2f9b from 42:09:78:65:a5:e5
842125 710.250071000 fe80::1805:6dff:fee… ff02::2 ICMPv6 70 Router Solicitation from 1a:05:6d:ef:d0:b2
842148 710.277705300 fe80::105e:1dff:fe2… ff02::16 ICMPv6 90 Multicast Listener Report Message v2
842218 710.307153000 fe80::4403:59ff:fe7… ff02::16 ICMPv6 90 Multicast Listener Report Message v2
842220 710.307154400 fe80::c056:aeff:fe8… ff02::16 ICMPv6 150 Multicast Listener Report Message v2
842234 710.310738700 :: ff02::1:ffd9:b041 ICMPv6 86 Neighbor Solicitation for fe80::5083:eaff:fed9:b041
842315 710.374151000 fe80::14ad:95ff:fe7… ff02::16 ICMPv6 90 Multicast Listener Report Message v2
842352 710.378981700 fe80::cf5b:edf9:d90… ff02::1:ff6d:8b45 ICMPv6 86 Neighbor Solicitation for fe80::b12a:52ac:7b6d:8b45 from 78:2b:46:74:0a:f4
842420 710.479625600 fe80::c050:adff:fee… ff02::1:ff2e:7ea0 ICMPv6 86 Neighbor Solicitation for fe80::105e:1dff:fe2e:7ea0 from c2:50:ad:e2:4f:55
842444 710.488464600 fe80::4403:59ff:fe7… ff02::16 ICMPv6 90 Multicast Listener Report Message v2
842459 710.496608500 fe80::9cb6:f3ff:feb… ff02::16 ICMPv6 110 Multicast Listener Report Message v2
842482 710.571974600 fe80::13a2:4937:7ab… ff02::16 ICMPv6 110 Multicast Listener Report Message v2
842492 710.577165000 fe80::9cf1:1eff:fef… ff02::16 ICMPv6 170 Multicast Listener Report Message v2
842507 710.582071200 fe80::2aad:18ff:fea… ff02::1:ff71:b8b6 ICMPv6 86 Neighbor Solicitation for fe80::14ad:95ff:fe71:b8b6 from 28:ad:18:a8:fd:c8
Frame 12: Packet, 86 bytes on wire (688 bits), 86 bytes captured (688 bits) on interface \Device\NPF_{F2F2918D-643E-49E2-9AA9-09C33633
Section number: 1
Interface id: 0 (\Device\NPF_{F2F2918D-643E-49E2-9AA9-09C3363374A2})
Encapsulation type: Ethernet (1)
Arrival Time: Nov 4, 2026 09:18:38.840616300 India Standard Time
UTC Arrival Time: Nov 4, 2026 03:48:38.840616300 UTC
Epoch Arrival Time: 1793764118.840616300
[Time shift for this packet: 0.000000000 seconds]
[Time delta from previous captured frame: 1.500 microseconds]
[Time since reference or first frame: 6.335000 milliseconds]
Frame Number: 12
Frame Length: 86 bytes (688 bits)
Capture Length: 86 bytes (688 bits)
[Frame is marked: False]
[Frame is ignored: False]
[Protocols in frame: eth:ethertype:ipv6:icmpv6]
Character encoding: ASCII (0)
[Coloring Rule Name: ICMP]
[Coloring Rule String: icmp || icmpv6]
Ethernet II, Src: WNC_f8:bf:1f (8c:53:e6:f8:bf:1f), Dst: IPv6mcast_ff:83:da:39 (33:33:ff:83:da:39)
0000 33 33 ff 83 da 39 8c 53 e6 f8 bf 1f 86 dd 60 00
0010 00 00 00 20 3a ff fe 80 00 00 00 00 00 00 a5 54
0020 52 86 3a 1c a3 b1 ff 02 00 00 00 00 00 00 00 00
0030 00 01 ff 83 da 39 87 00 48 9c 00 00 00 00 fe 80
0040 00 00 00 00 00 00 79 41 01 30 fc 83 da 39 01 01
0050 8c 53 e6 f8 bf 1f

4.tcp.len > 0



Inspect packet headers


Frame 1017: Packet, 518 bytes on wire (4144 bits), 518 bytes captured (4144 bits) on interface \Device\NPF_{F2F2918D-643E-49E2-9AA9-0
Section number: 1
Interface id: 0 (\Device\NPF_{F2F2918D-643E-49E2-9AA9-09C3363374A2})
Encapsulation type: Ethernet (1)
Arrival Time: Nov 4, 2026 09:18:39.728757800 India Standard Time
UTC Arrival Time: Nov 4, 2026 03:48:39.728757800 UTC
Epoch Arrival Time: 1793764119.728757800
[Time shift for this packet: 0.000000000 seconds]
[Time delta from previous captured frame: 9.385000 milliseconds]
[Time delta from previous displayed frame: 9.385000 milliseconds]
[Time since reference or first frame: 894.476500 milliseconds]
Frame Number: 1017
Frame Length: 518 bytes (4144 bits)
Capture Length: 518 bytes (4144 bits)
[Frame is marked: False]
[Frame is ignored: False]
[Protocols in frame: eth:ethertype:ip:tcp:tls]
Character encoding: ASCII (0)
[Coloring Rule Name: TCP]
[Coloring Rule String: tcp]
Ethernet II, Src: AzureWaveTec_be:92:f3 (00:e9:3a:be:92:f3), Dst: HewlettPacka_52:c1:00 (4c:d5:87:52:c1:00)
Destination: HewlettPacka_52:c1:00 (4c:d5:87:52:c1:00)
.... ..0. .... = LG bit: Globally unique address (factory default)
....0 .... = IG bit: Individual address (unicast)
Source: AzureWaveTec_be:92:f3 (00:e9:3a:be:92:f3)
.... ..0. .... = LG bit: Globally unique address (factory default)
....0 .... = IG bit: Individual address (unicast)
Type: IPv4 (0x0800)
[Stream index: 502]
Internet Protocol Version 4, Src: 10.250.22.44, Dst: 20.207.73.85
Transmission Control Protocol, Src Port: 53083, Dst Port: 443, Seq: 1, Ack: 1, Len: 464
Source Port: 53083
Destination Port: 443
[Stream index: 0]
[Stream Packet Number: 4]